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ADVANCED DATABASES WK5

1. SCENARIO ANALYSIS:

A common transaction scenario occurs when a customer uses mobile banking to transfer money from their bank account to another person. For example, a student may need to pay school fees and transfers the required amount from their bank account to the college's account. The transaction begins when the student logs into the mobile banking application and enters the recipient's account number, amount, and payment reference. The bank then verifies the customer's identity using a password, PIN, or one-time password (OTP). After verification, the bank checks whether sufficient funds are available in the account. If the information is correct and enough money is available, the transaction is processed and the amount is transferred to the college account. Both the sender and receiver receive a confirmation notification. Security measures such as encryption, authentication, and transaction monitoring help protect the transaction from fraud and unauthorized access. This process makes banking faster, convenient, and more secure.

2.CONCEPT RESEARCH:

Isolation levels are database settings that control how transactions interact with each other when multiple transactions run at the same time. They help maintain data consistency and prevent problems such as dirty reads, non-repeatable reads, and phantom reads. There are four common isolation levels: Read Uncommitted, Read Committed, Repeatable Read, and Serializable. Read Uncommitted provides the lowest level of protection but allows dirty reads. Read Committed prevents dirty reads, while Repeatable Read also prevents changes to data already read during a transaction. Serializable provides the highest isolation by making transactions behave as if they were executed one at a time. Isolation levels balance data accuracy, consistency, performance, and concurrency in database systems.

3.TOOL PRACTICE:

SQL Transaction:

```
START TRANSACTION;
```

```
UPDATE Accounts  
SET Balance = Balance - 500  
WHERE AccountID = 101;
```

```
UPDATE Accounts  
SET Balance = Balance + 500  
WHERE AccountID = 102;
```

COMMIT;

This SQL transaction demonstrates how money can be transferred safely between two bank accounts. The transaction subtracts 500 from the sender's account and adds 500 to the receiver's account. Both operations are treated as one unit of work. If both operations are successful, the COMMIT statement permanently saves the changes. If an error occurs, the transaction can be cancelled using ROLLBACK, preventing incomplete updates. Practicing transactions helped me understand how databases maintain accuracy and consistency when handling important operations such as banking and money transfers.

4.DIAGRAM:



